

07 Specification / 기술 명세

CPF Public API·SPI·Annotation·Config·State·Data·Runtime 의 정확한 계약을 빠르게 찾기 위한 기준 Reference 입니다.

CPF 공식 문서 · 기준 Source bd5fcd7707cf · 2026-08-15

문서 역할 정확한 이름·Signature·Default·State·Validation·Owner·경계를 확인하는 Reference 입니다. 사용 절차는 02/03/06, 운영 절차는 04/05 가 담당합니다.

CPF 전체 아키텍처



그림 1. CPF 전체 구성과 주요 실행 경로

업무 도메인은 CPF 공통 계약과 선택형 Starter 를 기반으로 구성되며, 외부 진입은 필요에 따라 Direct, L4/Ingress, Gateway 조합을 선택할 수 있습니다. Batch Runtime 은 대량·예약 실행을, ADM/BZA 는 각각 플랫폼 운영과 선택형 업무관리를 담당합니다. 아래 Specification 은 이 구조를 구성하는 API·SPI·설정·상태·데이터·Runtime 계약을 항목별로 정의합니다.

1. 기준 Stack·공식 DB Vendor

항목	기준
Java	25
Gradle	9.1.0
Spring Boot	4.1.0
Spring Cloud	2025.1.2
Spring Batch	6.0.4
DB Vendor	Oracle / PostgreSQL / MariaDB

2. Module Ownership

Module	Owner 책임	Public/Runtime 경계
cpf-core	Context·Result/Error·Domain Contract	Topology-independent Public 계약
cpf-starters	기능별 Public API·Provider·AutoConfiguration	선택 Dependency
cpf-admin	플랫폼 운영·관리	ADM Backend/Frontend
cpf-biz-admin	업무관리	선택형 BZA Runtime
cpf-batch	Batch API·Scheduler·Worker·Agent·Center-Cut	Batch Runtime
cpf-gateway	Route·Security·Traffic·Target	선택형 Gateway Runtime
cpf-tools	Generator·DB·Build·Verification	Tooling
cpf-education	교육·사용 검증	Sample/Consumer

3. Base Class API

Class	Method	Return/의미
CpfBaseController	requireCurrentContext()	CpfContext; 현재 Context 필수
CpfBaseController	requireContext()	CpfContextSnapshot
CpfBaseController	requireText(value, fieldName)	trim 문자열; null/blank 거부
CpfBaseController	requireValue(value, fieldName)	null 거부
CpfBaseController	requireRule(condition, message)	규칙 위반 시 Validation 예외
CpfBaseController	ok(body)	HTTP 200
CpfBaseController	created(location, body)	HTTP 201 + Location 검증
CpfBaseController	noContent()	HTTP 204
CpfBaseController	executionFacts(operation)	operation/transactionId/executionId/actorId/tenantId
CpfBaseService	requireText(value, fieldName)	Framework Service Base 의 최소 공통 검증
CpfBaseDao	requireCurrentContext()/requireContext()	DB 실행 Context
CpfBaseDao	statementId(namespace, statement)	namespace.statement
CpfBaseDao	boundedSize(requested, default, max)	0 이하→default, max 상한
CpfBaseDao	executionFacts(operation)	비민감 DB 실행 식별자

Generated Domain Base Service 는 Framework Base 위에 Domain 별 `requireTransactionId()`, `transactionSequence()`, `actorId()` 같은 공통 정책을 추가하는 구조입니다.

4. Persistence API

Contract	Method	계약
CpfDataOperations	selectList(operationId, parameter)	Provider-neutral Named 목록 조회
CpfDataOperations	selectOne(operationId, parameter)	Named 단건 조회
CpfDataOperations	insert/update/delete	영향 건수 int
CpfDataOperations	inRollbackOnlyTransaction(callback)	항상 Enclosing TX rollback-only
CpfJdbcOperations	queryList(sql, parameters, resultType)	Infrastructure Adapter 목록
CpfJdbcOperations	queryOne(...)	Infrastructure Adapter 단건
CpfJdbcOperations	update(...)	JDBC 변경
CpfJdbcOperations	inRollbackOnlyTransaction(...)	Rollback-only 검증

SQL 경계 Generated Business Source 는 Provider 타입이나 SQL Literal 을 분산시키지 않고 선택 Provider 와 CPF Vendor Catalog 경계를 사용합니다.

5. @CpfTx Annotation Specification

Attribute	Default/필수	계약
id	필수	안정 Transaction ID
name	필수	표시명
ownerDomain	필수	Owner Domain
transactionManager	""	Spring TransactionManager Bean
propagation	REQUIRED	Spring Propagation
isolation	DEFAULT	Spring Isolation
requiredPermission	""	Permission
auditReasonRequired	false	Reason 요구
visibility	PUBLIC	노출 정책
gatewayAllowed	true	Gateway 진입 허용
directAllowed	false	Direct 진입 허용
readOnly	false	조회 전용
timeoutSeconds	-1	Provider Default Timeout

Propagation	의미/주의
REQUIRED	기존 TX 참여, 없으면 생성
REQUIRES_NEW	기존 TX 중단 후 독립 TX; 별도 Bean Proxy 경계 권장
SUPPORTS	TX 있으면 참여
MANDATORY	상위 TX 필수
NOT_SUPPORTED	TX 밖에서 실행
NEVER	TX 가 있으면 오류
NESTED	Savepoint 기반; Provider 지원 확인

6. CpfResult Specification

Factory/Method	계약
success(data)	SUCCESS 결과 생성
businessFailure(code,message)	BUSINESS_FAILURE
technicalFailure(code,message)	TECHNICAL_FAILURE
unknown(code,message,recoveryInfo)	UNKNOWN + 복구 정보
isSuccess()	SUCCESS 판정
isBusinessFailure()	업무 실패 판정
isTechnicalFailure()	기술 실패 판정
isUnknown()	결과 불명 판정
fold(...)	상태별 함수형 분기
requireData()	SUCCESS Data 요구

7. Domain Client Specification

항목	계약
Public Interface	CpfDomainClient<I extends CpfRequest, O extends CpfResponse>
Method	CpfResult<O> execute(I request)
업무 Golden Path	Generated Typed Domain Client.execute(request)
Binding	AUTO / LOCAL / REMOTE
Router	Typed Client 뒤 Runtime Binding 계층; 일반 업무 직접 호출 대상 아님
Result	SUCCESS / BUSINESS_FAILURE / TECHNICAL_FAILURE / UNKNOWN

8. 주요 Annotation Catalog

Annotation	목적	대표 Attribute/Default
@CpfController	Controller 표준	실제 Source Target/Context 계약
@CpfService	Service 표준	Service Component 계약
@CpfRepository / @CpfDao	Persistence	DB 계층 식별
@CpfTx	Transaction	propagation REQUIRED, readOnly false, timeout -1
@CpfClient	외부 Client	system·operation·sideEffecting·contextRequired
@CpfTimeout	Timeout	millis
@CpfRetry	Retry	maxAttempts·delayMillis·reconcileUnknownOutcome
@CpfIdempotent	멱등성	operation·ttlSeconds·inProgressTimeoutSeconds·replayResult
@CpfMessageListener	Message Consumer	destination·consumerGroup·idempotencyRequired·contextRequired
@CpfPermission	권한	value[]·all
@CpfApprovalRequired	승인	action·approvals·reasonRequired
@CpfAudit	감사	action·reasonRequired
@CpfBatchJob	Batch Job	value; restartable=true; maxConcurrentExecutions=1
@CpfBatchStep	Batch Step	value; order=0; idempotent=true

9. Batch API Specification

Contract	정확한 계약
BatchExecutionTopology	LOCAL, PARALLEL_STEPS, LOCAL_PARTITION, REMOTE_PARTITION, REMOTE_CHUNK, REMOTE_STEP
BatchControlState	RESERVED, STARTING, STARTED, STOPPING, STOPPED, COMPLETED, FAILED, UNKNOWN_RESULT, ABANDONING, ABANDONED, REJECTED
CpfBatchOperationType	RUN, SCHEDULE_RUN, RESTART, RERUN, RETRY, STOP
BatchStepHandler.Status	COMPLETED, FAILED, RETRYABLE_FAILURE, UNKNOWN_RESULT, STOPPED

@CpfBatchJob	Default/계약
value	필수 Job ID
restartable	true
maxConcurrentExecutions	1

@CpfBatchStep	Default/계약
value	필수 Step ID
order	0
idempotent	true

10. CpfBatchExecutionRequest Specification

10.1 Fields

Field	Type/의미
jobId	String; 신규 실행 Job ID
scheduleId	String; 예약 실행 ID
sourceExecutionId	Long; Retry/Restart/Rerun/Stop 기준
jobParameters	String JSON; blank→{}
requestUser	String; 요청자
reason	String; 운영 사유
operationType	CpfBatchOperationType
lockRequired	boolean; 중복 실행 Lock
standardBatchId	String; On-demand 표준 ID
businessDate	String; 업무 기준일
idempotencyKey	String; On-demand 멍등 Key

10.2 Factory Methods

Factory	Operation/특징
run(...)	RUN, lock=true
scheduledRun(...)	SCHEDULE_RUN, lock=true
retry(...)	RETRY, sourceExecutionId
restart(...)	RESTART, 같은 JobInstance 재개
rerun(...)	RERUN, 신규 JobInstance
stop(...)	STOP, lock=false
onDemand(...)	RUN + standardBatchId/businessDate/idempotencyKey

11. BatchParameterDefinition Specification

Field	계약
name	필수 Token; 영문 시작, 최대 100 자
type	자원 Type Canonicalization
required/defaultValue	필수/기본값
identifying	JobInstance 식별 여부
sensitive	정규화 결과 Masking
label/description	운영 표시
allowedValues	ENUM 은 반드시 1 개 이상
pattern	Regex
min/maxValue	숫자 범위
min/maxLength	문자 길이
referenceType	Reference 유형
alias	표시/호환 별칭
overrideAllowed	Override 허용

지원 Type	지원 Type	지원 Type
STRING	INTEGER	LONG
DECIMAL	BOOLEAN	DATE
DATETIME	ENUM	JSON_OBJECT
SECRET_REFERENCE	PATH_ALIAS	SERVICE_REFERENCE
FILE_REFERENCE	CODE_REFERENCE	-

12. BatchExecutionControlPort

Method	반환	계약
start(BatchApprovedLaunchRequest)	BatchExecutionLink	승인된 실행 시작
stop(jobExecutionId, operatorId, reason)	boolean	정상 중지 요청
restart(jobExecutionId, operatorId, reason, fencingToken)	BatchExecutionLink	Checkpoint 재개
abandon(jobExecutionId, operatorId, reason)	void	Restart 대상 제외
reconcile(cpfExecutionId)	BatchExecutionLink	Control/실제 결과 대사

13. Batch Execution Configuration

Property	Default	Range
cpf.batch.execution.default-chunk-size	100	1..10,000
cpf.batch.execution.max-partition-count	256	1..10,000
cpf.batch.execution.remote-poll-interval-ms	1,000	10..60,000
cpf.batch.execution.remote-timeout-ms	3,600,000	1,000..86,400,000
cpf.batch.execution.remote-chunk-max-wait-timeouts	40	1..10,000
cpf.batch.execution.remote-chunk-throttle-limit	256	1..10,000
cpf.batch.execution.max-materialized-jobs	1,024	16..100,000
cpf.batch.execution.executor-core-size	4	1..512
cpf.batch.execution.executor-max-size	32	core..2,048
cpf.batch.execution.executor-queue-capacity	1,024	1..1,000,000

14. Gateway Route Specification

Field 그룹	Field	계약
식별	routeId, serviceId, operationId, standardExecutionId	routeId/serviceId 필수
Match	environmentCode, hostPattern, pathPattern, httpMethod, apiVersion	pathPattern 필수; 기본 DEFAULT/* /v1
Target	serverGroupId, endpoint(targetPath), ingressProtocol, targetProtocol	Enabled Route 는 serverGroupId 필수
Security	tlsPolicyId, authenticationPolicyId, authorizationPolicyId, headerPolicyId	TLS ingress 는 tlsPolicyId 필수
Traffic	rateLimitPolicyId, healthPolicyId, failoverGroupId	정책 참조
Timeout	connectTimeoutMs, responseTimeoutMs, overallTimeoutMs	양수; overall 은 각 timeout 이상
Retry	maxRetryCount, idempotent	Retry>0 이면 idempotent=true
Lifecycle	routeVersion, enabled, expectedVersion	Version/Optimistic Control

15. Gateway Registry·Target·Rate API

Contract	주요 Method/Field	계약
CpfGatewayRegistryPort	findServerGroups/findMembers/findBindings/ findApplyStatuses	Control Plane 조회
CpfGatewayRegistryPort	saveServerGroup/saveBinding/changeBindingState/delete*	Version·Reason 기반 변경
CpfGatewayRegistryPort	acknowledge	Instance expected/applied Version 결과
CpfGatewayRegistryPort	request/find/cancel/revalidateConnectionTest	Connection Test Lifecycle
CpfGatewayRegistryPort	claimHealthProbe(s)/reportHealth	Lease/Fencing Health
CpfGatewayTargetSelectionPort	select(SelectionRequest)	routable Candidate 선택
TargetCandidate	enabled/draining/maintenance/health/circuit/ activeRequests/latency/canary	Route 가능성 계산
CpfGatewayRateLimitPort	acquire(Request)	API/CLIENT/CHANNEL/TENANT Scope
Rate Request	permits 1..1000; 필수 execution/route/request ID	200 자 제약
Rate Decision	allowed/remaining/resetAt/retryAfter/duplicate/degraded/ reason	Caller 판정

16. Gateway Safety Configuration

Property	Default	Validation/영향
cpf.gateway.route-refresh	30s	positive
cpf.gateway.policy-refresh	15s	positive
cpf.gateway.health-probe-interval	30s	positive
cpf.gateway.stale-after	90s	positive
cpf.gateway.connect-timeout-cap	10s	positive
cpf.gateway.response-timeout-cap	60s	positive
cpf.gateway.overall-timeout-cap	90s	connect/response 이상
cpf.gateway.retry-count-cap	2	0..10
cpf.gateway.request-body-bytes-cap	10 MiB	0 이상
cpf.gateway.response-body-bytes-cap	10 MiB	0 이상
cpf.gateway.header-count-cap	100	≥1
cpf.gateway.header-bytes-cap	64 KiB	≥1024
cpf.gateway.bootstrap-mode	FAIL_CLOSED	FAIL_CLOSED/LAST_KNOWN_GOOD
cpf.gateway.rate-limit-counter-mode	LOCAL	LOCAL/JDBC
cpf.gateway.max-concurrent-requests-cap	1,000	1..1,000,000
cpf.gateway.require-tls-targets	true	Target TLS
cpf.gateway.allowed-target-ports	443,8443,9443	1..65535
cpf.gateway.maintenance-retry-after	60s	cap 이하

17. Gateway Worker Configuration

Property	Default	Consumer
cpf.gateway.route-refresh-millis	30000 ms	Route Snapshot
cpf.gateway.route-event.worker-millis	1000 ms	Route Event Worker
cpf.gateway.route-reconcile-millis	30000 ms	Route Reconcile
cpf.gateway.health.worker-millis	10000 ms	Health Worker
cpf.gateway.connection-test.worker-millis	2000 ms	Connection Test Worker

18. Security Technical Contract

영역	기술 계약	문서 경계
Authentication	User/Service Principal·Resource Server/OIDC 등 선택 Provider	정확한 Runtime Provider 는 구성별 확인
Authorization	Permission/Role/Operation	Fail-closed 권한 판단
Data Scope	조회/변경 범위	업무/운영 기능별 적용
Masking	Log/UI/Export 민감정보 보호	원문 노출 금지
Approval	위험 Action 승인	상태/승인 수/만료
Reason	위험 Action 사유	Audit 연계
Audit	Actor/Action/Target/Before/After/Result	민감정보 제외
Secret	Reference/External Provider	Source·Evidence 원문 금지

19. DB Lifecycle Specification

단계	Canonical 범위	검증
Install	빈 DB Object/Index/FK/Metadata	3 Vendor Renderer
Migration	Version/Checksum 순서	중복/Drift
Seed	기준정보	역등성
Verify	Table/Column/Index/Constraint/Sentinel	실제 Query
Upgrade	App/Schema 호환	Expand/Contract 영향
Rollback	지원 범위·데이터 손실·이전 Version	Vendor 별 SQL
Vendor	Oracle / PostgreSQL / MariaDB	비공식 Vendor 표기 금지

정보 경로	역할
cpf-tools/db/canonical/**	DB Canonical Source
cpf-tools/db/generated/current/oracle/**	Oracle Renderer
cpf-tools/db/generated/current/postgresql/**	PostgreSQL Renderer
cpf-tools/db/generated/current/mariadb/**	MariaDB Renderer
cpf-tools/db/generated/domain-template/**	Generated Domain DB Template

20. Runtime·Topology Contract

영역	계약
Domain	Same JVM Local / 분리 Runtime Remote / AUTO
Gateway	Direct·L4/Ingress·Gateway·L4+Gateway 선택
Batch	LOCAL·PARALLEL_STEPS·LOCAL/REMOTE_PARTITION·REMOTE_CHUNK/STEP
Multi-instance	Instance identity·Health·Version/Checksum·Lease/Fencing 등 Runtime 계약
Observability	Log·Metric·Trace·Health·Audit
UNKNOWN	결과 미확정; Status/Reconcile/Compensation 판단

21. Error·State·Recovery Contract

구분	정확한 의미	재시도/복구
BUSINESS_FAILURE	업무 거절 확정	일반적으로 입력/상태 수정
TECHNICAL_FAILURE	기술 실패 확정	역등성·정책하 Retry
UNKNOWN	실제 처리 여부 미확정	즉시 새 요청 제한; Reconcile
Retry	같은 처리 재시도	일시 오류
Restart	실행 상태 기반 재개	Batch Checkpoint
Reprocess	선택 업무 건 신규 처리	부분 실패
Reconcile	서로 다른 결과 대사	UNKNOWN/Partial
Compensation	확정 Side Effect 보정	분산 경계
Rollback	Local TX/Config/Artifact 원복	원복 가능한 경계

22. ADM·BZA·운영 UI Contract

UI 영역	정본	기술 확인
ADM Route	cpf-admin/frontend/src/app/routes.ts	Route·Risk·expectedOperationIds
BZA Route	cpf-biz-admin/frontend/src/app/routes.ts	Menu·Operation
Batch 운영	ADM batch-* routes	Execution/Scheduler/Worker/Recovery
Gateway 운영	ADM gateway-* routes	Route/Health/Apply/Transaction
Audit	ADM Audit Operation	Actor/Action/Result

23. Source Reference Index

계약	Source
CpfBaseController	cpf-starters/web/src/main/java/com/cpf/web/api/CpfBaseController.java
CpfBaseService	cpf-starters/base/runtime/src/main/java/com/cpf/foundation/api/CpfBaseService.java
CpfBaseDao	cpf-starters/data/persistence/src/main/java/com/cpf/data/persistence/api/CpfBaseDao.java
CpfTx	cpf-starters/data/persistence/src/main/java/com/cpf/data/persistence/api/annotation/CpfTx.java
CpfDomainClient	cpf-core/src/main/java/com/cpf/core/api/domain/CpfDomainClient.java
CpfResult	cpf-core/src/main/java/com/cpf/core/api/result/CpfResult.java
CpfDataOperations	cpf-starters/data/persistence/src/main/java/com/cpf/data/persistence/api/database/CpfDataOperations.java
CpfJdbcOperations	cpf-starters/data/persistence/src/main/java/com/cpf/data/persistence/api/database/CpfJdbcOperations.java
BatchParameterDefinition	cpf-batch/api/src/main/java/com/cpf/batch/api/BatchParameterDefinition.java
CpfBatchExecutionRequest	cpf-batch/api/src/main/java/com/cpf/batch/api/CpfBatchExecutionRequest.java
BatchExecutionControlPort	cpf-batch/api/src/main/java/com/cpf/batch/api/BatchExecutionControlPort.java
CpfBatchJob	cpf-batch/api/src/main/java/com/cpf/batch/api/annotation/CpfBatchJob.java
CpfBatchStep	cpf-batch/api/src/main/java/com/cpf/batch/api/annotation/CpfBatchStep.java
BatchStepHandler	cpf-batch/api/src/main/java/com/cpf/batch/spi/BatchStepHandler.java
CpfGatewayRoute	cpf-gateway/src/main/java/com/cpf/gateway/api/CpfGatewayRoute.java
CpfGatewayRegistryPort	cpf-gateway/src/main/java/com/cpf/gateway/api/CpfGatewayRegistryPort.java
CpfGatewayEntryPolicyPort	cpf-gateway/src/main/java/com/cpf/gateway/api/CpfGatewayEntryPolicyPort.java
CpfGatewayTargetSelectionPort	cpf-gateway/src/main/java/com/cpf/gateway/api/CpfGatewayTargetSelectionPort.java
CpfGatewayAuthenticationPort	cpf-gateway/src/main/java/com/cpf/gateway/api/CpfGatewayAuthenticationPort.java
CpfGatewayAuthorizationPort	cpf-gateway/src/main/java/com/cpf/gateway/api/CpfGatewayAuthorizationPort.java
CpfGatewayRateLimitPort	cpf-gateway/src/main/java/com/cpf/gateway/api/CpfGatewayRateLimitPort.java
CpfGatewaySafetyProperties	cpf-gateway/src/main/java/com/cpf/gateway/config/CpfGatewaySafetyProperties.java
Batch Execution Properties	cpf-batch/runtime/src/main/java/com/cpf/batch/execution/CpfBatchExecutionProperties.java
ADM Route	cpf-admin/frontend/src/app/routes.ts
BZA Route	cpf-biz-admin/frontend/src/app/routes.ts
Generator	cpf-tools/generator/**
DB	cpf-tools/db/**

계약	Source
Stack	gradle/cpf-stack.properties

24. 검증·제한 원칙

항목	판정 원칙
구현	Public API+Implementation+Consumer 를 함께 확인
Config	Property Type·Default·Validation·Consumer 확인
DB	Source/Renderer/Migration/Test 구분
Topology	구현 존재와 Runtime 검증 완료 구분
Test	미실행 Test 를 PASS 로 기록하지 않음
Deprecated	실제 Deprecated 가 있을 때 대체 계약 기록
지원 Vendor	Oracle/PostgreSQL/MariaDB 만 공식 범위
문서	Source Link 는 설명의 대체물이 아님

25. 공식 문서 책임 경계

질문	Owner 문서
CPF 로 일반 업무를 어떻게 개발?	02 프레임워크 개발자 가이드
Batch Job 을 어떻게 설계·개발?	03 배치 개발자 가이드
Runtime/ADM/BZA/Gateway 를 어떻게 운영?	04 운영자 매뉴얼
Batch 를 어떻게 실행·복구?	05 배치 운영 가이드
Gateway Route/Policy 를 어떻게 개발·적용?	06 Gateway 개발·사용 가이드
정확한 API/Property/State 값은?	07 Specification / 기술 명세